 

# KURE, Japan

WMO: **477660**

Lat: **34.2411N** Lon: **132.5503E** Elev: **5** StdP: **101.26** Time Zone: **9.00 (E09)** Period: **94-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **1** | **0.0** | **1.0** | **-8.1** | **1.9** | **4.4** | **-6.8** | **2.1** | **4.3** | **8.8** | **7.2** | **7.9** | **7.4** | **2.9** | **40** | **0.361** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **8** | **5.8** | **32.6** | **25.3** | **31.7** | **25.1** | **30.8** | **24.8** | **26.2** | **30.7** | **25.7** | **30.1** | **25.3** | **29.5** | **3.5** | **240** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **24.9** | **19.9** | **28.7** | **24.4** | **19.4** | **28.4** | **24.0** | **18.9** | **28.2** | **81.3** | **30.8** | **79.3** | **30.3** | **77.6** | **29.7** | **28.2** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **7.8** | **6.5** | **5.6** | DB | **-1.8** | **34.0** | **1.4** | **0.9** | **-2.8** | **34.6** | **-3.6** | **35.1** | **-4.4** | **35.6** | **-5.4** | **36.2** | (4) |
| (5) | | | | WB | **-3.6** | **26.7** | **1.3** | **0.6** | **-4.5** | **27.2** | **-5.3** | **27.6** | **-6.0** | **27.9** | **-6.9** | **28.4** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | | DBAvg | **16.7** | **6.0** | **6.5** | **9.6** | **14.5** | **19.2** | **22.7** | **26.9** | **28.3** | **24.8** | **19.5** | **13.6** | **8.4** | (6) |
| (7) | | | DBStd | **8.04** | **2.22** | **2.63** | **2.86** | **2.82** | **2.03** | **1.85** | **2.11** | **1.63** | **2.43** | **2.64** | **2.96** | **2.70** | (7) |
| (8) | | | HDD10.0 | **342** | **127** | **102** | **43** | **3** | **0** | **0** | **0** | **0** | **0** | **0** | **4** | **64** | (8) |
| (9) | | | HDD18.3 | **1586** | **384** | **330** | **271** | **118** | **14** | **0** | **0** | **0** | **0** | **18** | **142** | **308** | (9) |
| (10) | | | CDD10.0 | **2794** | **1** | **5** | **30** | **137** | **287** | **380** | **523** | **567** | **444** | **293** | **113** | **14** | (10) |
| (11) | | | CDD18.3 | **996** | **0** | **0** | **0** | **2** | **42** | **130** | **265** | **308** | **194** | **53** | **2** | **0** | (11) |
| (12) | | | CDH23.3 | **8285** | **0** | **0** | **0** | **5** | **103** | **538** | **2561** | **3488** | **1450** | **141** | **0** | **0** | (12) |
| (13) | | | CDH26.7 | **2527** | **0** | **0** | **0** | **0** | **4** | **59** | **822** | **1307** | **328** | **7** | **0** | **0** | (13) |
| (14) | Wind | | WSAvg | **2.3** | **2.7** | **2.5** | **2.5** | **2.3** | **2.1** | **1.9** | **2.2** | **2.3** | **2.3** | **2.3** | **2.4** | **2.7** | (14) |
| (15) | Precipitation | | PrecAvg | **1413** | **49** | **65** | **112** | **126** | **160** | **215** | **235** | **111** | **139** | **93** | **72** | **46** | (15) |
| (16) | | | PrecMax | **2172** | **144** | **117** | **228** | **214** | **345** | **481** | **583** | **314** | **248** | **264** | **151** | **104** | (16) |
| (17) | | | PrecMin | **829** | **9** | **26** | **48** | **36** | **38** | **38** | **33** | **13** | **50** | **16** | **16** | **3** | (17) |
| (18) | | | PrecStd | **308** | **30** | **27** | **43** | **47** | **69** | **105** | **159** | **85** | **64** | **70** | **41** | **26** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **13.8** | **16.6** | **19.4** | **23.9** | **27.1** | **29.7** | **33.5** | **34.0** | **31.9** | **27.7** | **22.1** | **16.9** | (19) |
| (20) | | | MCWB | | **10.1** | **12.2** | **13.3** | **15.8** | **17.9** | **23.0** | **25.6** | **25.6** | **24.8** | **22.2** | **16.8** | **12.3** | (20) |
| (21) | | 2% | DB | | **12.1** | **13.9** | **17.5** | **21.9** | **25.6** | **28.2** | **32.3** | **33.1** | **30.7** | **26.0** | **20.8** | **15.3** | (21) |
| (22) | | | MCWB | | **8.3** | **9.8** | **12.0** | **14.7** | **17.7** | **21.8** | **25.3** | **25.5** | **24.0** | **20.7** | **16.2** | **11.3** | (22) |
| (23) | | 5% | DB | | **11.0** | **12.2** | **16.0** | **20.5** | **24.4** | **27.2** | **31.4** | **32.2** | **29.6** | **24.9** | **19.7** | **14.1** | (23) |
| (24) | | | MCWB | | **7.3** | **8.5** | **11.2** | **14.3** | **17.3** | **21.2** | **25.0** | **25.3** | **23.5** | **19.6** | **15.5** | **10.2** | (24) |
| (25) | | 10% | DB | | **10.0** | **10.9** | **14.5** | **19.2** | **23.4** | **26.2** | **30.5** | **31.4** | **28.6** | **23.8** | **18.5** | **12.9** | (25) |
| (26) | | | MCWB | | **6.2** | **7.3** | **10.1** | **13.6** | **17.0** | **20.9** | **24.7** | **25.0** | **23.1** | **18.7** | **14.5** | **9.3** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **11.4** | **13.8** | **14.8** | **18.5** | **21.1** | **24.2** | **26.7** | **26.8** | **25.8** | **23.6** | **18.9** | **14.3** | (27) |
| (28) | | | MCDB | | **12.9** | **15.8** | **17.1** | **20.9** | **23.3** | **27.5** | **31.4** | **31.7** | **29.7** | **26.3** | **20.5** | **15.9** | (28) |
| (29) | | 2% | WB | | **9.2** | **10.9** | **13.2** | **16.9** | **20.0** | **23.6** | **26.1** | **26.3** | **25.1** | **22.2** | **17.5** | **12.1** | (29) |
| (30) | | | MCDB | | **11.3** | **13.1** | **15.9** | **19.4** | **22.5** | **26.5** | **30.5** | **31.0** | **28.8** | **25.0** | **19.7** | **14.2** | (30) |
| (31) | | 5% | WB | | **7.9** | **9.3** | **12.1** | **15.9** | **19.1** | **23.0** | **25.5** | **25.9** | **24.6** | **20.8** | **16.3** | **10.9** | (31) |
| (32) | | | MCDB | | **10.2** | **11.4** | **14.9** | **18.8** | **22.1** | **25.7** | **29.8** | **30.5** | **28.2** | **23.8** | **19.0** | **13.4** | (32) |
| (33) | | 10% | WB | | **6.7** | **7.9** | **10.9** | **14.9** | **18.2** | **22.2** | **25.1** | **25.6** | **24.0** | **19.4** | **15.0** | **9.7** | (33) |
| (34) | | | MCDB | | **9.5** | **10.6** | **14.1** | **18.1** | **21.7** | **24.9** | **29.3** | **30.0** | **27.5** | **22.9** | **18.1** | **12.6** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **6.2** | **6.6** | **7.3** | **7.5** | **7.4** | **5.9** | **5.4** | **5.8** | **5.9** | **6.5** | **6.7** | **6.4** | (35) |
| (36) | | 5% DB | MCDBR | | **7.4** | **8.1** | **9.0** | **9.1** | **9.0** | **7.4** | **6.6** | **6.7** | **6.5** | **7.0** | **7.3** | **7.5** | (36) |
| (37) | | | MCWBR | | **5.4** | **5.7** | **6.0** | **4.9** | **4.2** | **3.1** | **2.4** | **2.3** | **2.8** | **4.1** | **5.0** | **5.3** | (37) |
| (38) | | 5% WB | MCDBR | | **6.8** | **7.0** | **7.6** | **7.0** | **6.6** | **5.1** | **6.1** | **6.1** | **5.4** | **5.7** | **6.3** | **6.7** | (38) |
| (39) | | | MCWBR | | **5.6** | **5.7** | **6.0** | **4.6** | **4.0** | **2.6** | **2.3** | **2.3** | **2.5** | **4.0** | **5.0** | **5.5** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.405** | **0.449** | **0.511** | **0.540** | **0.541** | **0.559** | **0.559** | **0.529** | **0.488** | **0.442** | **0.418** | **0.399** | (40) |
| (41) | | taud | | | **2.140** | **2.045** | **1.898** | **1.880** | **1.943** | **1.977** | **2.029** | **2.113** | **2.188** | **2.221** | **2.192** | **2.174** | (41) |
| (42) | | Ebn at Noon | | | **773** | **779** | **763** | **763** | **768** | **751** | **748** | **765** | **777** | **780** | **755** | **753** | (42) |
| (43) | | Edh at Noon | | | **121** | **148** | **186** | **198** | **188** | **182** | **172** | **155** | **138** | **123** | **115** | **111** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **2.23** | **2.89** | **3.92** | **4.80** | **5.31** | **4.61** | **4.91** | **5.08** | **4.08** | **3.40** | **2.49** | **1.95** | (44) |
| (45) | | RadStd | | | **0.17** | **0.24** | **0.31** | **0.44** | **0.53** | **0.49** | **0.66** | **0.49** | **0.41** | **0.33** | **0.20** | **0.14** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (32 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page